

# Лабораторная работа №16

Базовая защита от атак типа «brute force» (Fail2ban)

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## Цель работы

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Получить практические навыки установки, настройки и проверки работы **Fail2ban** для защиты сервера от атак типа *brute force*.

## Установка Fail2ban

---

```
Installed:
  fail2ban-1.1.0-6.el10_0.noarch      fail2ban-firewalld-1.1.0-6.el10_0.noarch fail2ban-selinux-1.1.0-6.el10_0.noarch
  fail2ban-sendmail-1.1.0-6.el10_0.noarch fail2ban-server-1.1.0-6.el10_0.noarch

Complete!
[root@server.vabazlov.net server]#
[root@server.vabazlov.net server]# systemctl start fail2ban.service
[root@server.vabazlov.net server]# systemctl enable fail2ban.service
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' → '/usr/lib/systemd/system/fail2ban.service'.
[root@server.vabazlov.net server]# █
```

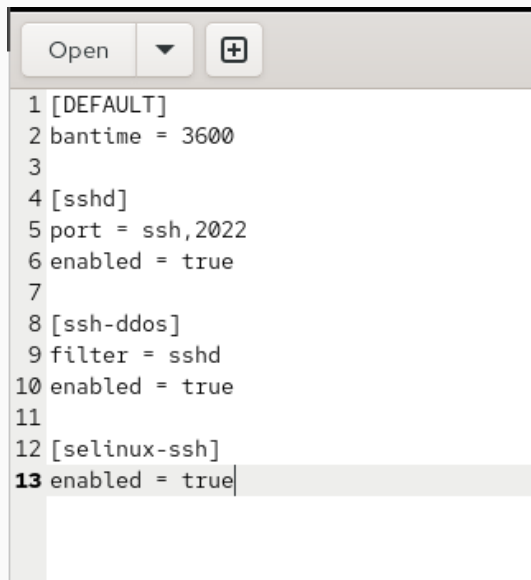
Рис. 1: Установка fail2ban

```
[vabazlov@server.vabazlov.net ~]$ sudo tail -f /var/log/fail2ban.log
[sudo] password for vabazlov:
2025-12-09 08:03:49,726 fail2ban.server [17258]: INFO -----
2025-12-09 08:03:49,726 fail2ban.server [17258]: INFO Starting Fail2ban v1.1.0
2025-12-09 08:03:49,726 fail2ban.observer [17258]: INFO Observer start...
2025-12-09 08:03:49,729 fail2ban.database [17258]: INFO Connected to fail2ban persistent database '/var/lib/fail2
ban/fail2ban.sqlite3'
2025-12-09 08:03:49,730 fail2ban.database [17258]: WARNING New database created. Version '4'
```

Рис. 2: Логи запуска

## Начальная конфигурация

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A screenshot of a configuration file editor window. The window has a title bar with three buttons: "Open", a dropdown arrow, and a "+" icon. The main area contains a text file with the following content:

```
1 [DEFAULT]
2 bantime = 3600
3
4 [sshd]
5 port = ssh,2022
6 enabled = true
7
8 [ssh-ddos]
9 filter = sshd
10 enabled = true
11
12 [selinux-ssh]
13 enabled = true
```

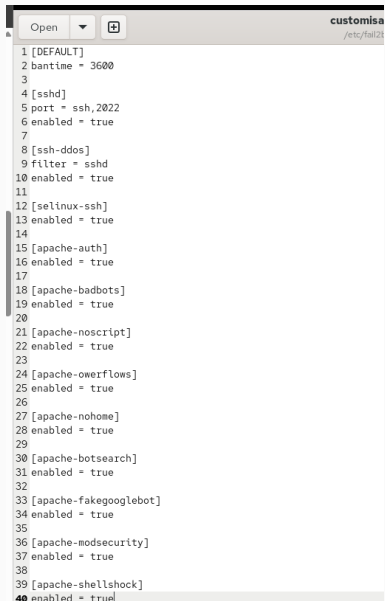
The line "13 enabled = true" is currently selected, highlighted in a light gray background.



## Применение и проверка

```
2025-12-09 08:06:49,550 fail2ban.server [17887]: INFO
2025-12-09 08:06:49,550 fail2ban.server [17887]: INFO Starting Fail2ban v1.1.0
2025-12-09 08:06:49,550 fail2ban.observer [17887]: INFO Observer start...
2025-12-09 08:06:49,555 fail2ban.database [17887]: INFO Connected to fail2ban persistent database '/var/lib/fail2
ban/fail2ban.sqlite3'
2025-12-09 08:06:49,556 fail2ban.jail [17887]: INFO Creating new jail 'sshd'
2025-12-09 08:06:49,562 fail2ban.jail [17887]: INFO Jail 'sshd' uses systemd {}
2025-12-09 08:06:49,563 fail2ban.jail [17887]: INFO Initiated 'systemd' backend
2025-12-09 08:06:49,563 fail2ban.filter [17887]: INFO maxLines: 1
2025-12-09 08:06:49,568 fail2ban.filtersystemd [17887]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.servi
ce + _COMM=sshd + _COMM=sshd-session'
2025-12-09 08:06:49,568 fail2ban.filter [17887]: INFO maxRetry: 5
2025-12-09 08:06:49,568 fail2ban.filter [17887]: INFO findtime: 600
2025-12-09 08:06:49,568 fail2ban.actions [17887]: INFO banTime: 3600
2025-12-09 08:06:49,568 fail2ban.filter [17887]: INFO encoding: UTF-8
2025-12-09 08:06:49,568 fail2ban.jail [17887]: INFO Creating new jail 'selinux-ssh'
2025-12-09 08:06:49,570 fail2ban.jail [17887]: INFO Jail 'selinux-ssh' uses pyinotify {}
2025-12-09 08:06:49,571 fail2ban.jail [17887]: INFO Initiated 'pyinotify' backend
2025-12-09 08:06:49,572 fail2ban.datedetector [17887]: INFO date pattern '': 'Epoch'
2025-12-09 08:06:49,572 fail2ban.filter [17887]: INFO maxRetry: 5
2025-12-09 08:06:49,572 fail2ban.filter [17887]: INFO findtime: 600
2025-12-09 08:06:49,572 fail2ban.actions [17887]: INFO banTime: 3600
2025-12-09 08:06:49,572 fail2ban.filter [17887]: INFO encoding: UTF-8
2025-12-09 08:06:49,572 fail2ban.filter [17887]: INFO Added logfile: '/var/log/audit/audit.log' (pos = 0, hash
= 8b32b1822504d7d10543d210ab694e54f1993c75)
2025-12-09 08:06:49,573 fail2ban.jail [17887]: INFO Creating new jail 'ssh-ddos'
2025-12-09 08:06:49,573 fail2ban.jail [17887]: INFO Jail 'ssh-ddos' uses pyinotify {}
2025-12-09 08:06:49,573 fail2ban.jail [17887]: INFO Initiated 'pyinotify' backend
2025-12-09 08:06:49,574 fail2ban.filter [17887]: INFO maxLines: 1
2025-12-09 08:06:49,574 fail2ban.filter [17887]: INFO maxRetry: 5
2025-12-09 08:06:49,574 fail2ban.filter [17887]: INFO findtime: 600
2025-12-09 08:06:49,574 fail2ban.actions [17887]: INFO banTime: 3600
2025-12-09 08:06:49,574 fail2ban.filter [17887]: INFO encoding: UTF-8
2025-12-09 08:06:49,575 fail2ban.jail [17887]: INFO Jail 'sshd' started
2025-12-09 08:06:49,575 fail2ban.jail [17887]: INFO Jail 'selinux-ssh' started
2025-12-09 08:06:49,575 fail2ban.jail [17887]: INFO Jail 'ssh-ddos' started
2025-12-09 08:06:49,578 fail2ban.filtersystemd [17887]: INFO [sshd] Jail is in operation now (process new journal entr
ies)
```

# Активирование модулей Apache



```
customisa
/etc/httpd/conf/httpd.conf

1 [DEFAULT]
2 bantime = 3600
3
4 [sshd]
5 port = ssh,2022
6 enabled = true
7
8 [ssh-ddos]
9 filter = sshd
10 enabled = true
11
12 [selinux-ssh]
13 enabled = true
14
15 [apache-auth]
16 enabled = true
17
18 [apache-badbots]
19 enabled = true
20
21 [apache-noscript]
22 enabled = true
23
24 [apache-overflow]
25 enabled = true
26
27 [apache-nohome]
28 enabled = true
29
30 [apache-botsearch]
31 enabled = true
32
33 [apache-fakegooglebot]
34 enabled = true
35
36 [apache-modsecurity]
37 enabled = true
38
39 [apache-shellshock]
40 enabled = true
```

## Проверка работы HTTP-защиты

```
2025-12-09 08:09:25,946 fail2ban.filter [18257]: INFO encoding: UTF-8
2025-12-09 08:09:25,946 fail2ban.filter [18257]: INFO Added logfile: '/var/log/httpd/server.vabazlov.net-error_
log' (pos = 0, hash = )
2025-12-09 08:09:25,946 fail2ban.filter [18257]: INFO Added logfile: '/var/log/httpd/error_log' (pos = 0, hash
= 26ccd59855c1f0441c21233de025777f217c5cee)
2025-12-09 08:09:25,946 fail2ban.filter [18257]: INFO Added logfile: '/var/log/httpd/ssl_error_log' (pos = 0, h
ash = 4907acf097a68318513ced05c89d9f76cf48b289)
2025-12-09 08:09:25,946 fail2ban.filter [18257]: INFO Added logfile: '/var/log/httpd/www.vabazlov.net-error_log
' (pos = 0, hash = 1eeaad48e05c6ce6fdc3cad3b479d6b71125874a)
2025-12-09 08:09:25,947 fail2ban.jail [18257]: INFO Creating new jail 'ssh-ddos'
2025-12-09 08:09:25,947 fail2ban.jail [18257]: INFO Jail 'ssh-ddos' uses pyinotify {}
2025-12-09 08:09:25,947 fail2ban.jail [18257]: INFO Initiated 'pyinotify' backend
2025-12-09 08:09:25,948 fail2ban.filter [18257]: INFO maxlines: 1
2025-12-09 08:09:25,948 fail2ban.filter [18257]: INFO maxRetry: 5
2025-12-09 08:09:25,948 fail2ban.filter [18257]: INFO findtime: 600
2025-12-09 08:09:25,948 fail2ban.actions [18257]: INFO banTime: 3600
2025-12-09 08:09:25,948 fail2ban.filter [18257]: INFO encoding: UTF-8
2025-12-09 08:09:25,948 fail2ban.transmitter [18257]: ERROR Jail 'apache-overflow' skipped, because of wrong configu
ration: Unable to read the filter 'apache-overflow'
2025-12-09 08:09:25,949 fail2ban.filtersystemd [18257]: INFO [sshd] Jail is in operation now (process new journal entr
ies)
2025-12-09 08:09:25,949 fail2ban.jail [18257]: INFO Jail 'sshd' started
2025-12-09 08:09:25,949 fail2ban.jail [18257]: INFO Jail 'selinux-ssh' started
2025-12-09 08:09:25,950 fail2ban.jail [18257]: INFO Jail 'apache-auth' started
2025-12-09 08:09:25,951 fail2ban.jail [18257]: INFO Jail 'apache-badbots' started
2025-12-09 08:09:25,952 fail2ban.jail [18257]: INFO Jail 'apache-noscript' started
2025-12-09 08:09:25,953 fail2ban.jail [18257]: INFO Jail 'apache-nohome' started
2025-12-09 08:09:25,953 fail2ban.jail [18257]: INFO Jail 'apache-botsearch' started
2025-12-09 08:09:25,954 fail2ban.jail [18257]: INFO Jail 'apache-fakegooglebot' started
2025-12-09 08:09:25,955 fail2ban.jail [18257]: INFO Jail 'apache-modsecurity' started
2025-12-09 08:09:25,956 fail2ban.jail [18257]: INFO Jail 'apache-shellshock' started
2025-12-09 08:09:25,956 fail2ban.jail [18257]: INFO Jail 'ssh-ddos' started
```

Рис. 6: Активация jail'ов

```
40 enabled = true
41
42 [postfix]
43 enabled = true
44
45 [postfix-rbl]
46 enabled = true
47
48 [dovecot]
49 enabled = true
50
51 [postfix-sasl]
52 enabled = true
```

Рис. 7: Почтовые jail'ы

```
2025-12-09 08:11:18,672 fail2ban.jail [18567]: INFO Jail 'sshd' started
2025-12-09 08:11:18,673 fail2ban.jail [18567]: INFO Jail 'selinux-ssh' started
2025-12-09 08:11:18,673 fail2ban.jail [18567]: INFO Jail 'apache-auth' started
2025-12-09 08:11:18,673 fail2ban.jail [18567]: INFO Jail 'apache-badbots' started
2025-12-09 08:11:18,674 fail2ban.jail [18567]: INFO Jail 'apache-noscript' started
2025-12-09 08:11:18,675 fail2ban.jail [18567]: INFO Jail 'apache-overflows' started
2025-12-09 08:11:18,675 fail2ban.jail [18567]: INFO Jail 'apache-nohome' started
2025-12-09 08:11:18,675 fail2ban.jail [18567]: INFO Jail 'apache-botsearch' started
2025-12-09 08:11:18,676 fail2ban.jail [18567]: INFO Jail 'apache-fakegooglebot' started
2025-12-09 08:11:18,677 fail2ban.jail [18567]: INFO Jail 'apache-modsecurity' started
2025-12-09 08:11:18,677 fail2ban.jail [18567]: INFO Jail 'apache-shellshock' started
2025-12-09 08:11:18,677 fail2ban.jail [18567]: INFO Jail 'postfix' started
2025-12-09 08:11:18,678 fail2ban.jail [18567]: INFO Jail 'postfix-rbl' started
2025-12-09 08:11:18,678 fail2ban.jail [18567]: INFO Jail 'dovecot' started
2025-12-09 08:11:18,679 fail2ban.filtersystemd [18567]: INFO [dovecot] Jail is in operation now (process new journal e
ntries)
2025-12-09 08:11:18,680 fail2ban.filtersystemd [18567]: INFO [postfix] Jail is in operation now (process new journal e
ntries)
2025-12-09 08:11:18,680 fail2ban.jail [18567]: INFO Jail 'postfix-sasl' started
2025-12-09 08:11:18,680 fail2ban.filtersystemd [18567]: INFO [postfix-sasl] Jail is in operation now (process new jour
nal entries)
2025-12-09 08:11:18,680 fail2ban.jail [18567]: INFO Jail 'ssh-ddos' started
2025-12-09 08:11:18,680 fail2ban.filtersystemd [18567]: INFO [postfix-rbl] Jail is in operation now (process new journa
l entries)
```

Рис. 8: Запуск всех jail'ов

## Проверка Fail2ban

---

```
[root@server.vabazlov.net server]#  
[root@server.vabazlov.net server]# fail2ban-client status  
Status  
|- Number of jail:      16  
`- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-modsecurity, apache-nohome, ap  
ache-noscript, apache-overflows, apache-shellshock, dovecot, postfix, postfix-rbl, postfix-sasl, selinux-ssh, ssh-ddos, ss  
hd  
[root@server.vabazlov.net server]# fail2ban-client status sshd  
Status for the jail: sshd  
|- Filter  
|  |- Currently failed: 0  
|  |- Total failed:     0  
|  `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session  
`- Actions  
   |- Currently banned: 0  
   |- Total banned:     0  
   `-- Banned IP list:  
[root@server.vabazlov.net server]# fail2ban-client set sshd maxretry 2  
2  
[root@server.vabazlov.net server]# █
```

Рис. 9: Статус Fail2ban

- уменьшено maxretry до 2;
- выполнены неверные попытки входа.

```
2025-12-09 08:12:32,693 fail2ban.filter [18567]: INFO maxRetry: 2
2025-12-09 08:13:21,493 fail2ban.filter [18567]: INFO [sshd] Found 192.168.1.30 - 2025-12-09 08:13:21
2025-12-09 08:13:25,459 fail2ban.filter [18567]: INFO [sshd] Found 192.168.1.30 - 2025-12-09 08:13:24
2025-12-09 08:13:25,560 fail2ban.actions [18567]: NOTICE [sshd] Ban 192.168.1.30
2025-12-09 08:13:28,949 fail2ban.filter [18567]: INFO [sshd] Found 192.168.1.30 - 2025-12-09 08:13:28
2025-12-09 08:14:06,431 fail2ban.transmitter [18567]: ERROR Command ['set', 'sshd', 'ubnanip', '192.168.1.30'] has fa
iled. Received Exception("Invalid command 'ubnanip' (no set action or not yet implemented)")
2025-12-09 08:14:20,192 fail2ban.actions [18567]: NOTICE [sshd] Unban 192.168.1.30
■
```

Рис. 10: Блокировка IP



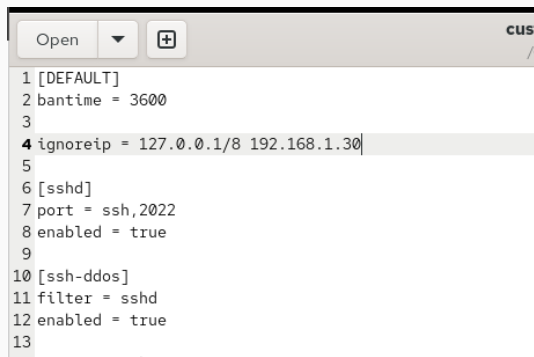
## Разблокировка и повторная проверка

```
[root@server.vabazlov.net server]#  
[root@server.vabazlov.net server]# fail2ban-client status sshd  
Status for the jail: sshd  
|- Filter  
| |- Currently failed: 1  
| |- Total failed: 3  
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session  
`- Actions  
  |- Currently banned: 1  
  |- Total banned: 1  
  `-- Banned IP list: 192.168.1.30  
[root@server.vabazlov.net server]# fail2ban-client set sshd unbanip 192.168.1.30  
2025-12-09 08:14:06,432 fail2ban [18970]: ERROR NOK: ("Invalid command 'unbanip' (no set action or not yet implemented)",)  
Invalid command 'unbanip' (no set action or not yet implemented)  
[root@server.vabazlov.net server]# fail2ban-client set sshd unbanip 192.168.1.30  
1  
[root@server.vabazlov.net server]# fail2ban-client status sshd  
Status for the jail: sshd  
|- Filter  
| |- Currently failed: 1  
| |- Total failed: 3  
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session  
`- Actions  
  |- Currently banned: 0  
  |- Total banned: 1  
  `-- Banned IP list:  
[root@server.vabazlov.net server]#
```

Рис. 11: Unban и проверка

## Игнорирование IP

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The image shows a configuration file editor window. The title bar includes an 'Open' button, a dropdown arrow, and a '+' icon. The file name 'cust' is visible on the right. The editor contains a configuration file with the following content:

```
1 [DEFAULT]
2 bantime = 3600
3
4 ignoreip = 127.0.0.1/8 192.168.1.30|
5
6 [sshd]
7 port = ssh,2022
8 enabled = true
9
10 [ssh-ddos]
11 filter = sshd
12 enabled = true
13
14 ...
```

Рис. 12: ignoreip

```
2025-12-09 08:16:04,471 fail2ban.filter      [19221]: INFO      [sshd] Ignore 192.168.1.30 by ip
2025-12-09 08:16:08,704 fail2ban.filter      [19221]: INFO      [sshd] Ignore 192.168.1.30 by ip
2025-12-09 08:16:11,449 fail2ban.filter      [19221]: INFO      [sshd] Ignore 192.168.1.30 by ip
```

Рис. 13: ignoreip в логах

```
[root@server.vabazlov.net server]#  
[root@server.vabazlov.net server]# fail2ban-client status sshd  
Status for the jail: sshd  
|- Filter  
| |- Currently failed: 0  
| |- Total failed: 0  
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session  
`- Actions  
  |- Currently banned: 0  
  |- Total banned: 0  
  `-- Banned IP list:  
[root@server.vabazlov.net server]#
```

Рис. 14: Нет блокировок

## Итоги работы

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В ходе лабораторной работы:

- установлен и настроен Fail2ban;
- активирована защита SSH, Apache и почтовых служб;
- проверен механизм блокировки и разблокировки IP;
- реализована автоматизация настройки Fail2ban в составе Vagrant-среды.